

AS Chemistry — Topic 1 Atomic Structure (Exam)

CIE 9701 AS Chemistry · 34 marks · about 40–45 minutes

Topic 1: particles and isotopes, mass spectrometry, electronic configuration / orbitals, and ionisation energy. Answer in English. Section A is Paper 1 style (1 mark each). Section B is written; show working and give the usual CIE reasons (nuclear charge, radius, shielding, spin-pair repulsion) where asked. A Periodic Table may be used.

Section	Content	Marks
A	Multiple choice	13
B1	Iron mass spectrum, particles and ionisation energy	11
B2	Electronic configuration, orbitals, ionisation energy and radii	10
Total		34

Section A — Multiple-choice questions (13 marks)**A1.**

[1]

The table shows the numbers of protons, neutrons and electrons in four different particles, W, X, Y and Z.

particle	number of protons	number of neutrons	number of electrons
W	32	40	32
X	32	40	34
Y	32	42	32
Z	34	40	34

Which pair represents the atoms of two isotopes of the same element?

A W and Y**B** W and Z**C** X and Y**D** X and Z**A2.**

[1]

An ion with a charge of $2-$ contains 10 electrons and 14 neutrons.

What is its nucleon number?

A 14**B** 22**C** 24**D** 26**A3.**

[1]

What is the total number of protons, neutrons and electrons present in an ammonium ion with a relative formula mass of 21?

	number of protons	number of neutrons	number of electrons
A	11	10	10
B	10	11	11
C	10	11	10
D	11	10	11

A4.

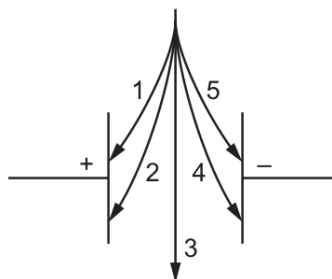
[1]

A helium ion contains two protons, two neutrons and one electron.

This helium ion and a proton are passed separately through a uniform electric field.

The particles are travelling at the same velocity.

Which arrow describes the path of each particle?



	helium ion	proton
A	1	2
B	2	1
C	4	5
D	5	4

A5.

[1]

A sample of magnesium contains the isotopes ^{24}Mg , ^{25}Mg and ^{26}Mg only.

The percentage abundance of ^{25}Mg and ^{26}Mg is the same.

The relative atomic mass of magnesium in the sample is 24.3.

What is the percentage abundance of ^{24}Mg ?

A 10%**B** 20%**C** 60%**D** 80%**A6.**

[1]

Which isolated gaseous atom has a total of five electrons occupying spherically shaped orbitals?

A sodium**B** fluorine**C** boron**D** potassium**A7.**

[1]

What is the electrons in boxes notation for the Fe^{3+} ion?

		3d				4s
A	[Ar]	↑↓	↑			↑↓
B	[Ar]	↑	↑	↑		↑↓
C	[Ar]	↑↓	↑	↑	↑	
D	[Ar]	↑	↑	↑	↑	

A8.

[1]

Which atom has more unpaired electrons than paired electrons in orbitals of principal quantum number 2?

A carbon**B** nitrogen**C** oxygen**D** fluorine**A9.**

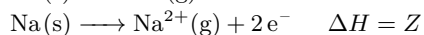
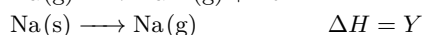
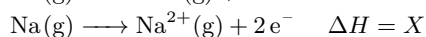
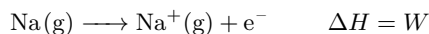
[1]

Why is the first ionisation energy of oxygen less than that of nitrogen?

A The nitrogen atom has its outer electron in a different subshell.**B** The nuclear charge on the oxygen atom is greater than that on the nitrogen atom.**C** The oxygen atom has a pair of electrons in one p orbital that repel one another.**D** There is more shielding in an oxygen atom.**A10.**

[1]

Equations involving four enthalpy changes are shown.



Which equation represents the second ionisation energy of sodium?

A X**B** $X + Y - W$ **C** $X - W$ **D** $Z - W$

A11.

[1]

The first seven ionisation energies of an element between lithium and neon in the Periodic Table are shown.

1310 3390 5320 7450 11000 13300 71000 kJ mol^{-1}

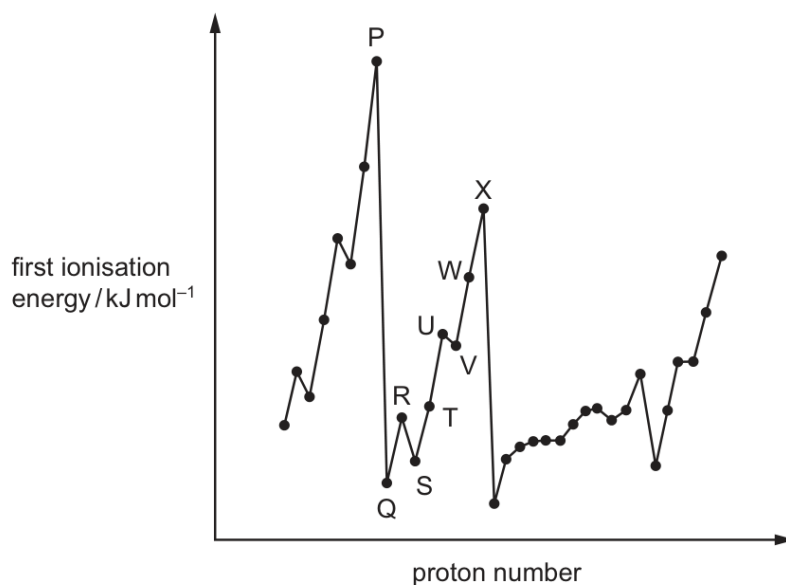
What is the outer electronic configuration of the element?

A $2s^2$ **B** $2s^2 2p^1$ **C** $2s^2 2p^4$ **D** $2s^2 2p^6$ **A12.**

[1]

The graph shows the variation of the first ionisation energy with proton number for some elements.

The letters used are **not** the actual symbols for the elements.



Which statement about the elements is correct?

- A** P and X are in the same period in the Periodic Table.
- B** The general increase from Q to X is due to increasing atomic radius.
- C** The small decrease from R to S is due to decreased shielding.
- D** The small decrease from U to V is due to repulsion between paired electrons.

A13.

[1]

Which statements about first ionisation energies are correct?

1. They are always endothermic.
2. They decrease down Group 2.
3. They decrease across Period 3.

Which statements are correct?

A	B	C	D
1, 2 and 3	1 and 2 only	2 and 3 only	1 only

Section B — Structured questions (21 marks)**B1 · Iron mass spectrum, particles and ionisation energy (11 marks)**

A sample of iron is analysed using a mass spectrometer. The mass spectrum shows three isotopes of iron are present in the sample.

B1(a)

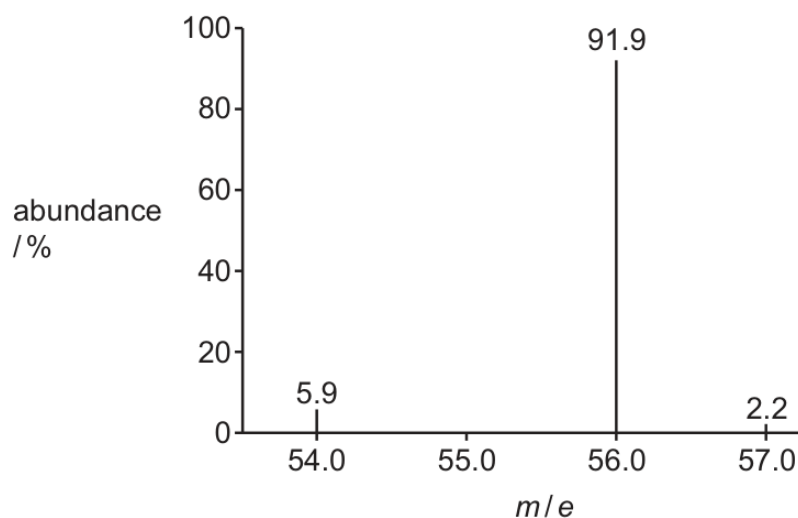
[1]

Define isotopes.

B1(b)(i)

[2]

Fig. 2.1 shows the mass spectrum of the sample of iron.

**Fig. 2.1**

Use Fig. 2.1 to calculate the relative atomic mass, A_r , of iron to one decimal place.

Show your working.

A_r of iron = _____

B1(b)(ii)

[2]

Complete Table 2.1 to show the number of protons and nucleons in one atom of ^{56}Fe .

particle	number of particles in one atom of ^{56}Fe
protons	
nucleons	

Table 2.1

B1(c)

[1]

Deduce the number of pairs of electrons in the shell with principal quantum number $n = 3$ in an Fe atom. number of pairs of electrons _____

B1(d)

[1]

Write an equation to represent the first ionisation energy of iron.

B1(e)

[4]

Suggest how the value for the first ionisation energy of ^{54}Fe compares to the first ionisation energy of ^{56}Fe . Explain your answer in terms of the factors that affect ionisation energy.

B2 · Electronic configuration, orbitals, ionisation energy and radii (10 marks)

The chemical properties of an element are related to the electronic configuration of its atoms.

B2(a)(i)

[1]

Give the full electronic configuration of a fluorine atom.

B2(a)(ii)

[1]

Deduce the number of pairs of electrons in the second energy shell, $n = 2$, of an oxygen atom. number of pairs of electrons

B2(a)(iii)

[1]

Draw the shape of the highest energy orbital that contains electrons in an atom of calcium.

**B2(b)(i)**

[1]

Write an equation to represent the first ionisation energy of sulfur.

B2(b)(ii)

[2]

Explain why the first ionisation energy of sulfur is less than the first ionisation energy of phosphorus.

B2(b)(iii)

[4]

Arrange the three species F^- , Ne and Na^+ in order of increasing radius.

Explain your answer.

_____ < _____ < _____
smallest radius largest radius